# 2-Colouring Shift-chains

Zak Smith

Andrzej Antoni Kwaśniewski

Tutor: Prof. Piotr Micek

November 8, 2026

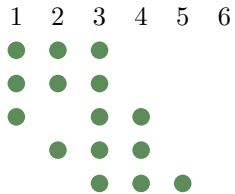


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Given an $r$-uniform hypergraph $H = (V, E)$ and a total ordering $<_V$ on $V$, each edge $e$ can be interpreted as a vector:

$$\vec{e} = \{ e_1, \ldots, e_r \}$$

where $e_1 < e_2 < \cdots < e_r$.




We define a partial order on edges by

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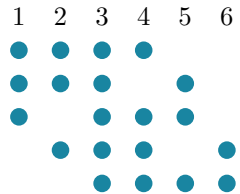
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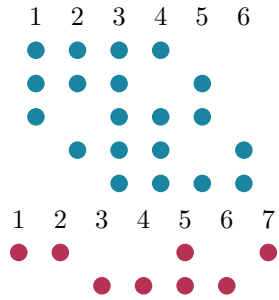
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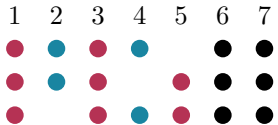
> **Theorem (Smith, 2026)**
>
> *All shift-chains of uniformity at least $4$ are properly 2-colourable.*

**Lemma**

*If every $r_0$-shift-chain is $k$ colourable, then for $r > r_0$, every $r$-shift-chain is $k$ colourable*

**Proof**

*We can just colour the first $r_0$ vertices with $k$ colours and colour the rest arbitrarily.*

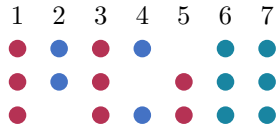


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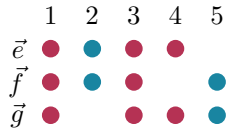
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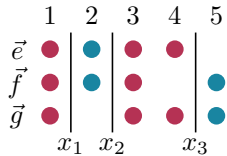
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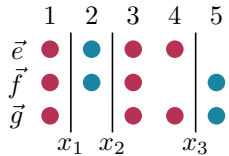
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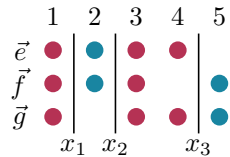
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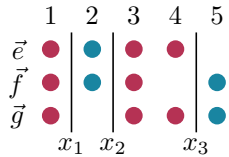



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Edge $\vec{e}$ contains switch $x$ if $e_i < x < e_{i+1}$

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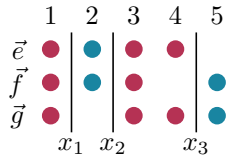
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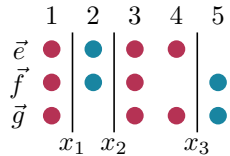



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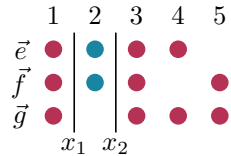



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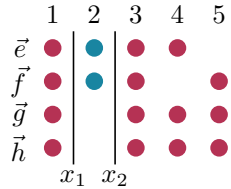
Colouring $c$ is $k$-well spread if every edge has switch count $1 \leq |S_c(\vec{e})| \leq k$

# Switch positions and monochromatic edges



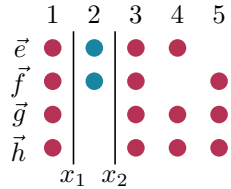

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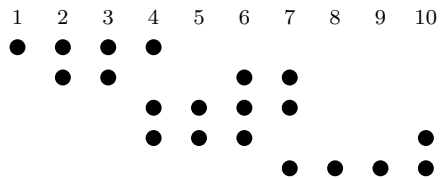
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- An edge is monochromatic iff every gap has an even number of switches.

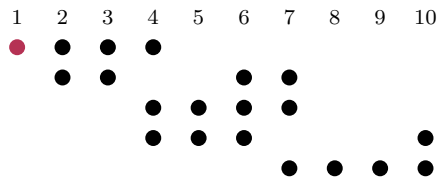
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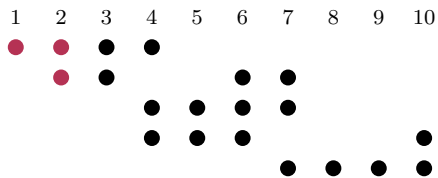
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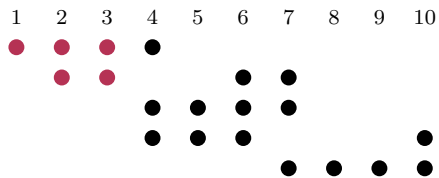
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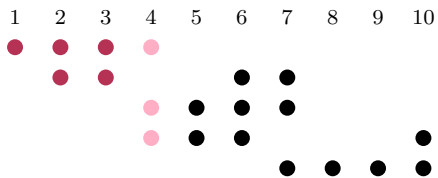
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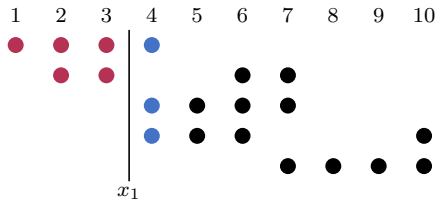
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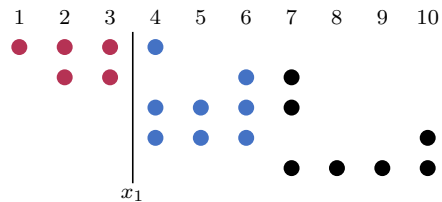
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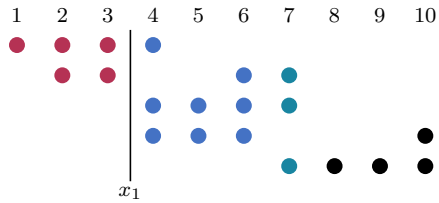
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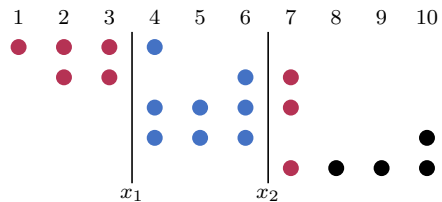
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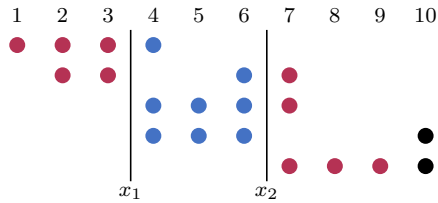
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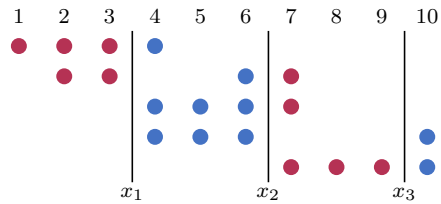
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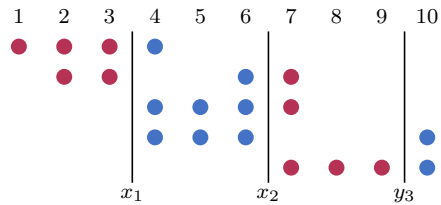
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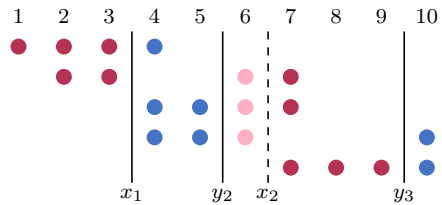
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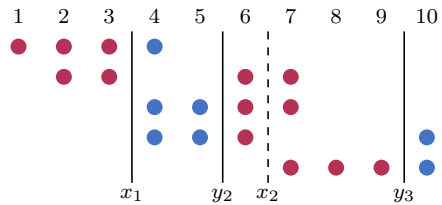
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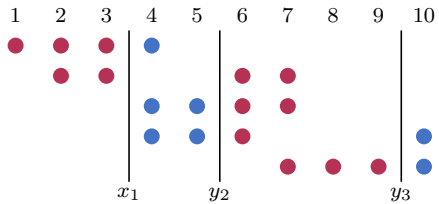


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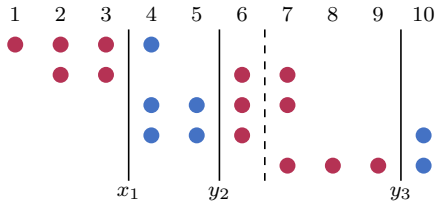


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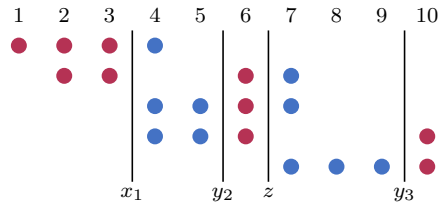


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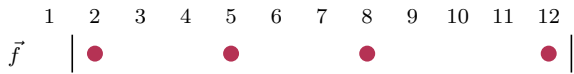
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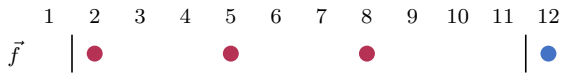


# X1: every edge contains one or two switches



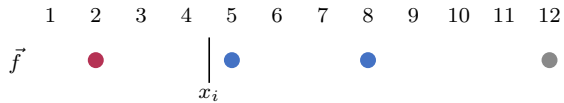

At least 1 switch by greedy colouring design

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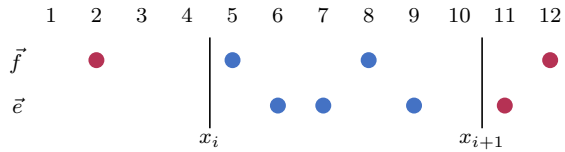

At least 1 switch by greedy colouring design

# X2: the second switch is immediately before $f_4$



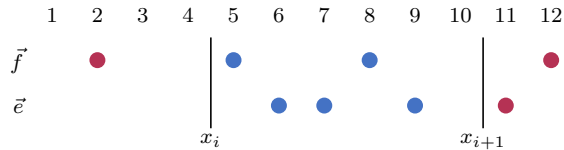

- The first switch $x_i$ lies inside $\vec{f}$.

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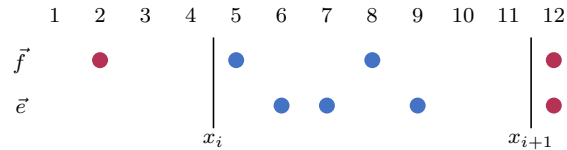

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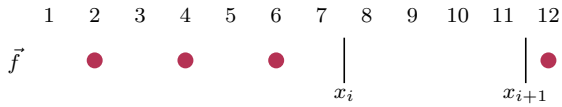
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- $x_{i+1} = e_4^- < f_4 \Rightarrow e_4 \leq f_4$. Thus $e_4 = f_4$ and $x_{i+1} = f_4^-$: no third switch.

# X3: monochromatic edges are $(3, 4)$-edges

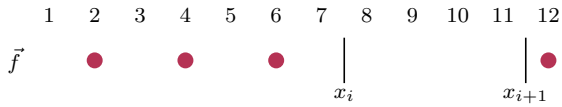
**X1.** Every edge $\vec{f}$ has one or two switches in $c_0$: $1 \leq |S_{c_0}(\vec{f})| \leq 2$.
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- A monochromatic edge has an even number of switches in each gap.

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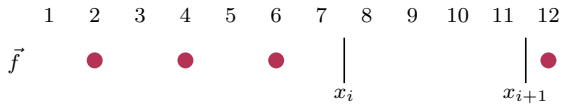
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- By X2, $x_{i+1} = f_4^-$, so $\vec{f}$ is a $(3, 4)$-edge.

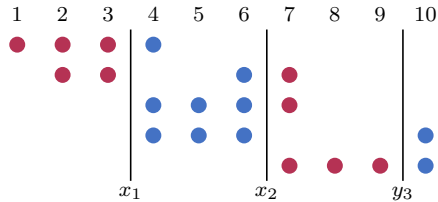
# Stage X: conclusions

- **(X1)** Every edge $\vec{f}$ has one or two switches: $1 \leq |S_{c_0}(\vec{f})| \leq 2$.
- **(X2)** If $\vec{f}$ contains $x_i$ and $x_{i+1}$, then $x_{i+1} = f_4^-$.
- **(X3)** Every monochromatic edge in $c_0$ is a $(3,4)$-edge.

# Stage Y: recap

**X3.** Every monochromatic edge in $c_0$ is a $(3, 4)$-edge.

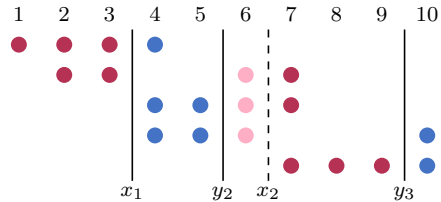
Stage X leaves only $(3, 4)$-edges monochromatic. We process them right to left, moving the first switch in each edge to just before its third vertex.



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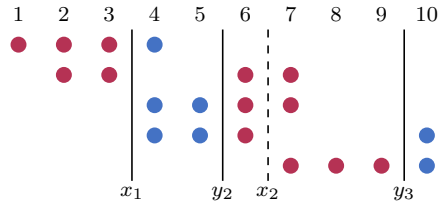
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# Y1, Y4: base case from X1 and X3

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We prove Y1–Y5 by induction on $t$.

**Claim**

**Y1.** *$c_t$ is 2-well-spread: every edge has one or two switches.*

**Y4.** *There are no $(1,2)$-edges in $c_t$.*

**Proof**

- *X1 gives Y1 at $t = 0$.*

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- *X1 gives Y1 at $t = 0$.*
- *X3 gives Y4 at $t = 0$.*

# Y2–Y3: base case from switch positions

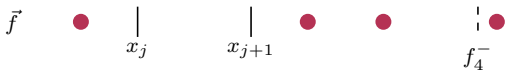
**X2.** If $\vec{f}$ contains $x_j, x_{j+1}$, then $x_{j+1} = f_4^-$.

## Claim

**Y2.** $x_{j-1} < y_j(t) \leq x_j$ *for* $j \in [s]$.
**Y3.** *For* $\vec{f} \in E(H)$ *and* $j \in [s]$, $f_1 < x_j \Rightarrow f_2 < y_{j+1}(t)$.

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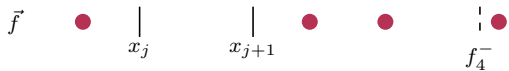
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- *Y2:* $y_j(0) = x_j$. *For Y3, suppose* $f_1 < x_j < x_{j+1} \leq f_2$.
- *Then* $\vec{f}$ *contains both switches. X2 forces* $x_{j+1} = f_4^- > f_2$*: impossible.*




*Hypothetical* $(1, 2)$*-edge: X2 would put the second bar just before* $f_4$.

# Y5: base case from X3

**X3.** Every monochromatic edge in $c_0$ is a $(3,4)$-edge.

## Claim

**Y5.** *If $\vec{f}$ is a $(2,3)$-edge with switches $y_{j-1}(t), y_j(t)$ for $j \in [2,s]$, then $f_3^- = y_j(t) < x_j$, and some $\vec{g} \in E(H)$ has $y_j(t) = g_3^-$ and $y_{j+1}(t) < g_4$.*

## Proof

- *X3 gives no $(2,3)$-edges at $t = 0$, so Y5 holds vacuously.*

# Y1–Y5: one Stage Y step

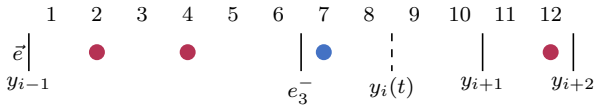
At time $t$, choose a maximal $(3,4)$-edge $\vec{e}$. Move only $y_i(t)$, setting $y_i(t+1) = e_3^-$.




- $y_{i-1}(t) < e_1 < e_3 < y_i(t) < y_{i+1}(t) < e_4 < y_{i+2}(t)$.

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# Y1–Y5: the moved edge starts after $x_{i-1}$

**Y2.** $x_{j-1} < y_j(t) \leq x_j$ for $j \in [s]$.
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**Claim**

$x_{i-1} < e_1$ *(automatic for* $i = 1$*).*

**Proof**

- *For* $i \geq 2$*, suppose* $e_1 < x_{i-1}$.

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- *X2 forces* $x_i = e_4^-$*, contradicting* $x_i < y_{i+1}(t) < e_4$*.*



# Y1: no edge loses its last switch

**Claim**

**Y1.** *$c_t$ is 2-well-spread: every edge has one or two switches.*

**Proof**

- *Suppose $\vec{f}$ has no switch in $c_{t+1}$. Since Y1 holds in $c_t$, its only old switch was $y_i(t)$.*

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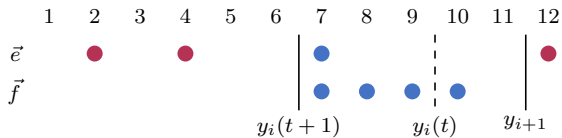
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- *Hence $y_i(t+1) < f_1 < y_i(t) < f_4 < y_{i+1}(t)$.*
- *But $e_1 < y_i(t+1) < f_1$ and $f_4 < y_{i+1}(t) < e_4$. Thus $e_1 < f_1$ and $f_4 < e_4$: $\vec{e}, \vec{f}$ are incomparable.*



# Y1: no edge gains a third switch

**Y2.** $x_{j-1} < y_j(t) \leq x_j$ for $j \in [s]$.
**X2.** If $\vec{f}$ contains $x_j, x_{j+1}$, then $x_{j+1} = f_4^-$.
**Auxiliary claim.** $x_{i-1} < e_1$.

## Claim

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- *Suppose $\vec{f}$ has three switches in $c_{t+1}$. The new one is $y_i(t+1) = e_3^-$.*

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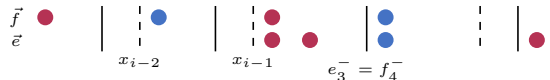
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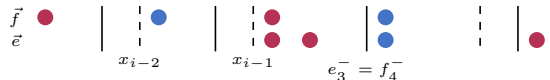
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# Y2–Y3: positions survive the move

**Auxiliary claim.** $x_{i-1} < e_1$.

## Claim

**Y2.** $x_{j-1} < y_j(t) \leq x_j$ *for* $j \in [s]$.
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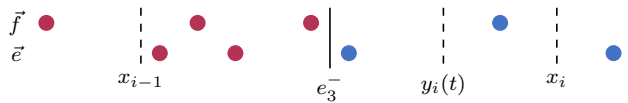
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- *Y3: only* $j = i-1$ *is new. If* $f_1 < x_{i-1} < e_1$, *then* $\vec{f} < \vec{e}$.




$f_1 < e_1 \implies f_2 \le e_2 < e_3^-$; *dashed bars mark reference positions.*

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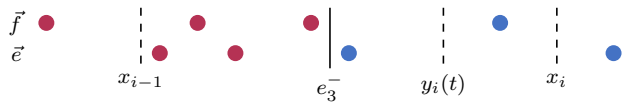
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# Y4: no $(1, 2)$-edge appears

**Y1.** $c_t$ is 2-well-spread: every edge has one or two switches.
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**Claim**

**Y4.** *There are no* $(1, 2)$*-edges in* $c_t$*.*

**Proof**

- *Suppose* $\vec{f}$ *has two switches* $y_j(t+1), y_{j+1}(t+1)$ *in* $(f_1, f_2)$*.*

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- *If* $j = i-1$*, then*

$$f_1 < y_{i-1}(t+1) < e_1 < e_2 < y_i(t+1) < f_2.$$

*So* $f_1 < e_1$ *but* $e_2 < f_2$*, contradicting comparability.*

1 2 3 4 5 6 7 8 9 10 11 12 13 14

# Y5: old $(2,3)$-edges keep their witness

**Y1.** $c_t$ is 2-well-spread: every edge has one or two switches.
**Y5.** If $\vec{f}$ is a $(2,3)$-edge with switches $y_{j-1}(t), y_j(t)$ for $j \in [2,s]$, then $f_3^- = y_j(t) < x_j$, and some $\vec{g} \in E(H)$ has $y_j(t) = g_3^-$ and $y_{j+1}(t) < g_4$.

## Claim

**Y5.** *If $\vec{f}$ is a $(2,3)$-edge with switches $y_{j-1}(t), y_j(t)$ for $j \in [2,s]$, then $f_3^- = y_j(t) < x_j$, and some $\vec{g} \in E(H)$ has $y_j(t) = g_3^-$ and $y_{j+1}(t) < g_4$.*

## Proof

- *If $\vec{f}$ misses the new switch, its two switches stay: losing the old one would leave at most one.*

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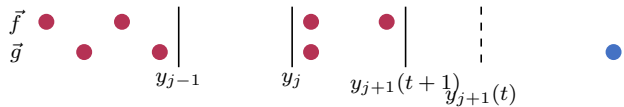
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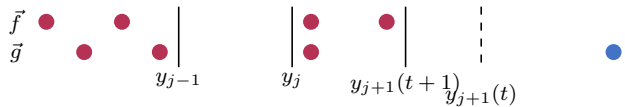
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## Proof

- *If $\vec{f}$ misses the new switch, its two switches stay: losing the old one would leave at most one.*
- *The old witness has $f_3^- = y_j = g_3^- < x_j$ and $y_{j+1}(t) < g_4$.*
- *Only $y_{j+1}$ might move left; still $y_{j+1}(t+1) \le y_{j+1}(t) < g_4$.*



# Y5: a new $(2,3)$-edge uses $y_{i-1}, y_i$

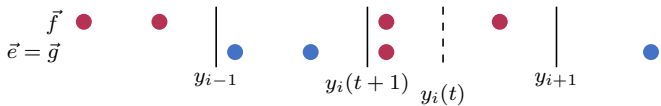
**Y1.** $c_t$ is 2-well-spread: every edge has one or two switches.
**Y2.** $x_{j-1} < y_j(t) \le x_j$ for $j \in [s]$.

## Claim

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## Proof

- *For this pair, $f_4 < y_{i+1} < e_4$ forces $\vec{f} < \vec{e}$, so $f_3 \le e_3$.*




*The two bounds pin both third vertices to the dot just after $y_i(t+1)$.*

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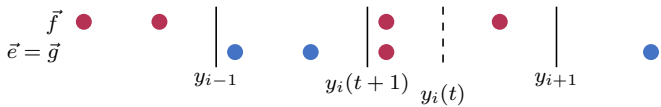
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## Proof

- *For this pair, $f_4 < y_{i+1} < e_4$ forces $\vec{f} < \vec{e}$, so $f_3 \le e_3$.*
- *But $e_3^- = y_i(t+1) < f_3$, so $e_3 \le f_3$. Thus $f_3 = e_3$.*




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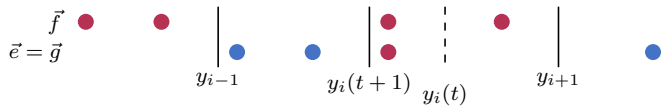
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## Proof

- *For this pair, $f_4 < y_{i+1} < e_4$ forces $\vec{f} < \vec{e}$, so $f_3 \le e_3$.*
- *But $e_3^- = y_i(t+1) < f_3$, so $e_3 \le f_3$. Thus $f_3 = e_3$.*
- *Take $\vec{g} = \vec{e}$: $f_3^- = y_i(t+1) < y_i(t) \le x_i$ and $y_{i+1} < g_4$.*




*The two bounds pin both third vertices to the dot just after $y_i(t+1)$.*

# Y5: the pair $y_i, y_{i+1}$ keeps its old witness

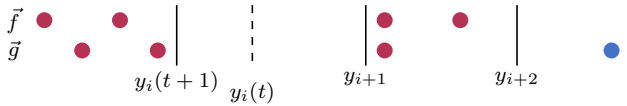
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## Proof

- $f_2 < y_i(t+1) < y_i(t) < y_{i+1} < f_3$: *both old switches were already in* $(f_2, f_3)$.



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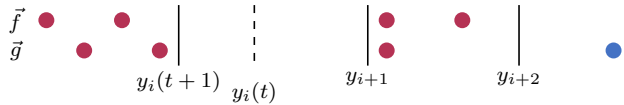
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## Proof

- $f_2 < y_i(t+1) < y_i(t) < y_{i+1} < f_3$*: both old switches were already in* $(f_2, f_3)$.
- *By Y1 these were the only old switches. Apply Y5 with* $j = i+1$.



# Y5: the pair $y_i, y_{i+1}$ keeps its old witness

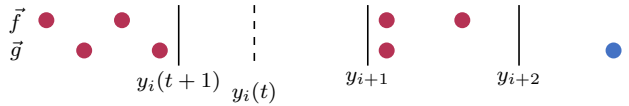
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- *By Y1 these were the only old switches. Apply Y5 with* $j = i+1$.
- $f_3^- = y_{i+1} < x_{i+1}$ *and the witness* $g_3 = f_3$, $y_{i+2} < g_4$ *remain.*



# Y6: chosen $(3, 4)$-edges move left

**Y1.** $c_t$ is 2-well-spread: every edge has one or two switches.

**Claim**

$\vec{e}(t+1) < \vec{e}(t)$.

**Proof**

- *Suppose the next chosen edge $\vec{f} > \vec{e}$. It must be new, since $\vec{e}$ was maximal.*

# Y6: chosen $(3,4)$-edges move left

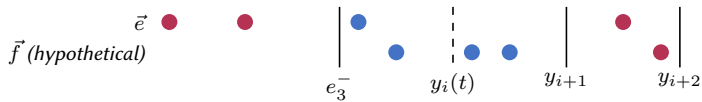
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- $y_i(t+1) = e_3^- < e_3 \le f_3$. *A $(3,4)$-edge has both switches after $f_3$, so the new bar is outside $\vec{f}$.*




*The moved bar lies left of $f_1$: it cannot supply a new pair in $(f_3, f_4)$.*

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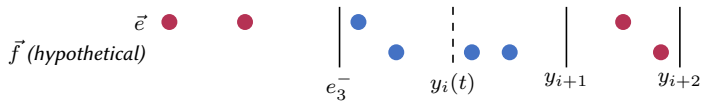
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- $y_i(t+1) = e_3^- < e_3 \le f_3$. *A $(3,4)$-edge has both switches after $f_3$, so the new bar is outside $\vec{f}$.*
- *If the old bar was outside too, $\vec{f}$ was already monochromatic. If inside, Y1 leaves at most one switch. Both are impossible.*




*The moved bar lies left of $f_1$: it cannot supply a new pair in $(f_3, f_4)$.*

# Y6: termination leaves only $(2, 3)$-edges

**Y1.** $c_t$ is 2-well-spread: every edge has one or two switches.
**Y4.** There are no $(1, 2)$-edges in $c_t$.
**Previous step.** $\vec{e}(t+1) < \vec{e}(t)$.

**Claim**

**Y6.** *Every monochromatic edge in $c_Y$ is a $(2, 3)$-edge.*

**Proof**

- *The indices cannot increase: otherwise* $e_4(t) < y_{i(t)+2} \leq y_{i(t+1)+1} < e_4(t+1)$.

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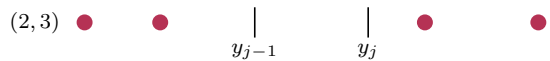
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- *Each move shifts a bar left; the moved indices never increase. The new walls strictly decrease: fewer than $n$ moves.*
- *At termination: no $(3,4)$-edge; Y4 excludes $(1,2)$-edges. By Y1, only $(2,3)$-edges can be monochromatic.*

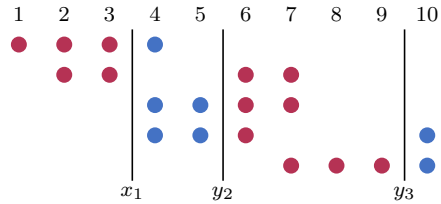



*The only surviving monochromatic pattern: two bars in the middle gap.*

# Stage Z: insert a switch after $e_3$

**Y6.** Every monochromatic edge in $c_Y$ is a $(2,3)$-edge.

Process the remaining $(2,3)$-edges left to right. Insert $z = e_3^+$ after the third vertex.

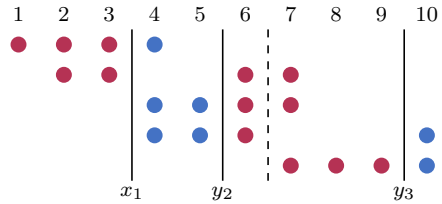



- A $(2,3)$-edge has two switches before its third vertex.

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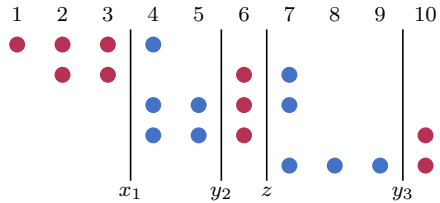



- A $(2,3)$-edge has two switches before its third vertex.
- Mark the new switch after that vertex.

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Process the remaining $(2,3)$-edges left to right. Insert $z = e_3^+$ after the third vertex.




- A $(2,3)$-edge has two switches before its third vertex.
- Mark the new switch after that vertex.
- Recolour the vertices crossed by $z$.

# Z1–Z2: base case at $\hat{c}_0 = c_Y$

**Y1.** $c_t$ is 2-well-spread: every edge has one or two switches.

## Claim

**Z1.** *$\hat{c}_t$ is 3-well-spread: every edge has at most three switches.*
**Z2.** $M(\hat{c}_t) \subsetneq M(\hat{c}_{t-1})$ *for* $t \geq 1$.

## Proof

- *At $t = 0$, Y1 gives one or two switches per edge, hence at most three.*




*Initial edge patterns already satisfy the bound of three.*

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## Proof

- *At $t = 0$, Y1 gives one or two switches per edge, hence at most three.*
- *Z2 concerns $t \geq 1$: there is no insertion to check yet.*




*Initial edge patterns already satisfy the bound of three.*

# Z3–Z4: base case at $\hat{c}_0 = c_Y$

**Y6.** Every monochromatic edge in $c_Y$ is a $(2,3)$-edge.

## Claim

**Z3.** *Every monochromatic edge in $\hat{c}_t$ is a $(2,3)$-edge, with $z < \vec{f}$ for all $z \in Z(t)$.*
**Z4.** *For each $z \in Z(t)$, some $j \in [s]$ and $v \in [n]$ satisfy $y_j = v^-$ and $z = v^+ \le x_j$.*

## Proof

- *Y6 gives the $(2,3)$ pattern for every remaining monochromatic edge.*




*Two $Y$-bars in the middle gap; no $Z$-bars yet.*

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## Proof

- *Y6 gives the $(2,3)$ pattern for every remaining monochromatic edge.*
- *$Z(0) = \varnothing$: both assertions about inserted switches hold vacuously.*

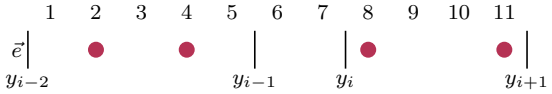



*Two $Y$-bars in the middle gap; no $Z$-bars yet.*

# Z1–Z4: insert the next switch

**Y5.** If $\vec{f}$ is a $(2,3)$-edge with switches $y_{j-1}(t), y_j(t)$ for $j \in [2,s]$, then $f_3^- = y_j(t) < x_j$, and some $\vec{g} \in E(H)$ has $y_j(t) = g_3^-$ and $y_{j+1}(t) < g_4$.

At time $t$, we have $\hat{c}_t$. Choose a minimal $(2,3)$-edge $\vec{e}$ and set $\hat{z} = e_3^+$.

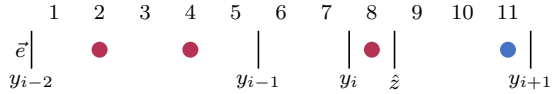



- $y_{i-2} < e_1 < e_2 < y_{i-1} < y_i < e_3 < e_4 < y_{i+1}$.

# Z1–Z4: insert the next switch

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At time $t$, we have $\hat{c}_t$. Choose a minimal $(2,3)$-edge $\vec{e}$ and set $\hat{z} = e_3^+$.




- $y_{i-2} < e_1 < e_2 < y_{i-1} < y_i < e_3 < e_4 < y_{i+1}$.
- Y5 gives $y_i = e_3^- < x_i$ and an edge $\vec{g}$ with $g_3 = e_3$ and $y_{i+1} < g_4$.

# Z1: every edge has at most three switches

**Y1.** $c_t$ is 2-well-spread: every edge has one or two switches.
**Z4.** For each $z \in Z(t)$, some $j \in [s]$ and $v \in [n]$ satisfy $y_j = v^-$ and $z = v^+ \leq x_j$.
**Z3.** Every monochromatic edge in $\hat{c}_t$ is a $(2,3)$-edge, with $z < \vec{f}$ for all $z \in Z(t)$.
**Y2.** $x_{j-1} < y_j(t) \leq x_j$ for $j \in [s]$.
**X2.** If $\vec{f}$ contains $x_j, x_{j+1}$, then $x_{j+1} = f_4^-$.

## Claim

**Z1.** *$\hat{c}_t$ is 3-well-spread: every edge has at most three switches.*

## Proof

- *A fourth switch would require an older $z \in Z(t)$ as well as $\hat{z}$: Y1 allows at most two $Y$-bars.*

# Z1: every edge has at most three switches

**Y1.** $c_t$ is 2-well-spread: every edge has one or two switches.
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- *Z4 and Z3 give $z = v^+ \le x_j$, $y_j = v^-$, and $z < e_1 < y_{i-1}$. Thus $j \le i-2$.*
- *$f_1 < z \le x_{i-2} < x_{i-1} < y_i < \hat{z} < f_4$. X2 forces $x_{i-1} = f_4^-$, impossible with $\hat{z}$ between them.*



# Z2: a new monochromatic edge has two switches

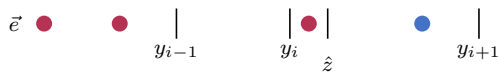
**Y1.** $c_t$ is 2-well-spread: every edge has one or two switches.
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## Claim

**Z2.** $M(\hat{c}_t) \subsetneq M(\hat{c}_{t-1})$ *for* $t \geq 1$.

## Proof

- *The chosen* $\vec{e}$ *is repaired:* $\hat{z} = e_3^+$ *separates its last two vertices.*




$\{y_i, \hat{z}\}$ *or* $\{\hat{z}, y_{i+1}\}$*: the two cases to exclude.*

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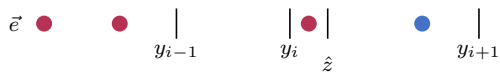
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## Proof

- *The chosen* $\vec{e}$ *is repaired:* $\hat{z} = e_3^+$ *separates its last two vertices.*
- *A new monochromatic* $\vec{f}$ *would have exactly two switches in one gap: Z1 bounds the count, Y1 excludes zero.*




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# Z2: a new monochromatic edge has two switches

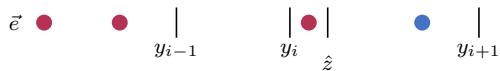
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## Claim

**Z2.** $M(\hat{c}_t) \subsetneq M(\hat{c}_{t-1})$ *for* $t \geq 1$.

## Proof

- *The chosen* $\vec{e}$ *is repaired:* $\hat{z} = e_3^+$ *separates its last two vertices.*
- *A new monochromatic* $\vec{f}$ *would have exactly two switches in one gap: Z1 bounds the count, Y1 excludes zero.*
- *One switch is* $\hat{z}$*. Its only possible partner is a neighbour,* $y_i$ *or* $y_{i+1}$*; all older* $z$ *lie before* $e_1$*.*




$\{y_i, \hat{z}\}$ *or* $\{\hat{z}, y_{i+1}\}$*: the two cases to exclude.*

# Z2: $y_i, \hat{z}$ cannot lie in the last gap

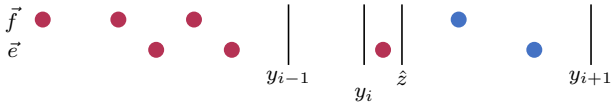
**Y5.** If $\vec{f}$ is a $(2,3)$-edge with switches $y_{j-1}(t), y_j(t)$ for $j \in [2,s]$, then $f_3^- = y_j(t) < x_j$, and some $\vec{g} \in E(H)$ has $y_j(t) = g_3^-$ and $y_{j+1}(t) < g_4$.

**Claim**

*No new monochromatic edge has switch pair $\{y_i, \hat{z}\}$ in $(f_3, f_4)$.*

**Proof**

- $f_3 < y_i < e_3$ *forces* $\vec{f} < \vec{e}$, *hence* $f_1 \leq e_1$.




*The forced extra bar changes the colour of $f_4$.*

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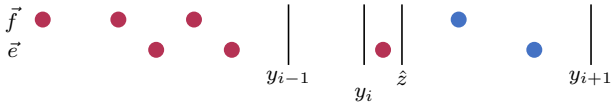
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## Proof

- $f_3 < y_i < e_3$ *forces* $\vec{f} < \vec{e}$*, hence* $f_1 \le e_1$.
- *But* $f_1 \le e_1 < y_{i-1} < \hat{z} < f_4$*: a third switch* $y_{i-1}$ *lies in* $\vec{f}$.




*The forced extra bar changes the colour of $f_4$.*

# Z2: $y_i, \hat{z}$ cannot lie in an earlier gap

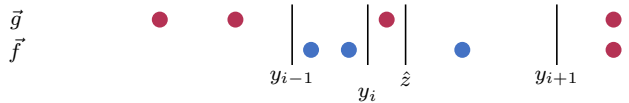
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**Claim**

*No new monochromatic edge has switch pair $\{y_i, \hat{z}\}$ in $(f_1, f_2)$ or $(f_2, f_3)$.*

**Proof**

- *The witness has $g_3 = e_3 < \hat{z} < f_3$, so $\vec{g} < \vec{f}$.*




*Middle-gap case shown; both earlier gaps force the witness endpoint past $y_{i+1}$.*

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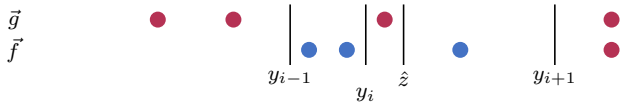
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**Proof**

- *The witness has $g_3 = e_3 < \hat{z} < f_3$, so $\vec{g} < \vec{f}$.*
- *Then $y_{i+1} < g_4 \le f_4$: a third switch $y_{i+1}$ lies in $\vec{f}$.*




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# Z2: the pair $\hat{z}, y_{i+1}$ is impossible

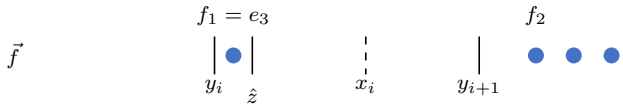
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*No new monochromatic edge has switch pair* $\{\hat{z}, y_{i+1}\}$.

**Proof**

- $y_i = e_3^- < f_1 < \hat{z} = e_3^+$ *forces* $f_1 = e_3$.



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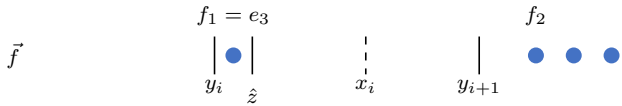
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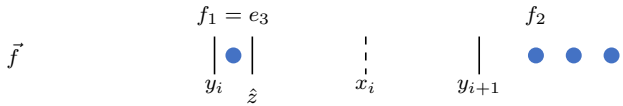
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**Proof**

- $y_i = e_3^- < f_1 < \hat{z} = e_3^+$ *forces* $f_1 = e_3$.
- *Both switches must lie in the first gap:* $y_{i+1} < f_2$.
- *Y5 gives* $f_1 < \hat{z} \leq x_i$*; Y3 gives* $f_2 < y_{i+1}$*, a contradiction.*



# Z3: every surviving edge starts after the new switch

**Z2.** $M(\hat{c}_t) \subsetneq M(\hat{c}_{t-1})$ for $t \geq 1$.
**Z3.** Every monochromatic edge in $\hat{c}_t$ is a $(2,3)$-edge, with $z < \vec{f}$ for all $z \in Z(t)$.

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**Proof**

- *By Z2, a surviving monochromatic $\vec{f}$ was already a $(2,3)$-edge.*




*The repaired edge has two colours; the surviving $(2,3)$-edge lies to its right.*

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## Proof

- *By Z2, a surviving monochromatic $\vec{f}$ was already a $(2,3)$-edge.*
- *It cannot contain $\hat{z}$: three switches cannot leave all four vertices the same colour.*
- *Minimality gives $\vec{e} < \vec{f}$, so $\hat{z} < e_4 \leq f_4$. Being outside $\vec{f}$, the new switch must lie before $f_1$.*




*The repaired edge has two colours; the surviving $(2,3)$-edge lies to its right.*

# Z4: the new switch brackets the third vertex

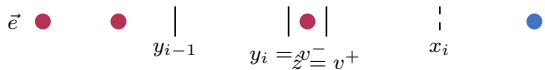
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## Proof

- *Y5 gives $y_i = e_3^- < x_i$. The next wall is $e_3^+$, hence $\hat{z} = e_3^+ \le x_i$.*




*The two adjacent walls bracket $v = e_3$; the right wall is still at or before $x_i$.*

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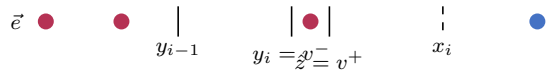
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## Proof

- *Y5 gives $y_i = e_3^- < x_i$. The next wall is $e_3^+$, hence $\hat{z} = e_3^+ \le x_i$.*
- *Take $v = e_3$ and $j = i$. Every older inserted switch keeps its witness.*




*The two adjacent walls bracket $v = e_3$; the right wall is still at or before $x_i$.*

# Z1–Z4: termination gives a proper 2-colouring

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**Claim**

*$c_{Y \cup Z}$ is a proper 2-colouring.*

**Proof**

- *Each step inserts a distinct wall, so at most $n - 1$ insertions are possible.*

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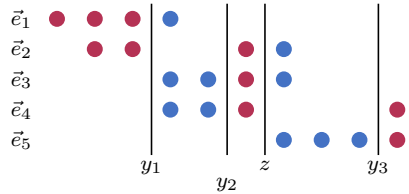
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**Claim**

*$c_{Y \cup Z}$ is a proper 2-colouring.*

**Proof**

- *Each step inserts a distinct wall, so at most $n-1$ insertions are possible.*
- *At termination no $(2,3)$-edge remains. By Z3, no monochromatic edge remains.*




*The opening example after Stage Z: every row contains both colours.*

# Theorem: every $r \geq 4$ shift-chain is 2-colourable

**X1.** Every edge $\vec{f}$ has one or two switches in $c_0$: $1 \leq |S_{c_0}(\vec{f})| \leq 2$.
**X3.** Every monochromatic edge in $c_0$ is a $(3,4)$-edge.
**Y6.** Every monochromatic edge in $c_Y$ is a $(2,3)$-edge.
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- **Z:** Insert switches; the monochromatic-edge set strictly shrinks to empty.

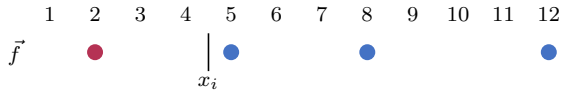
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- **Y:** Move switches left; monochromatic edges can only be $(2,3)$-edges.
- **Z:** Insert switches; the monochromatic-edge set strictly shrinks to empty.
- This proves the 4-uniform case. The first-four-vertices reduction gives every uniformity $r \geq 4$.

# X1: one or two switches give a 3-colouring

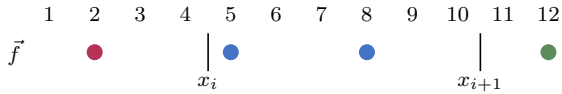
**X1.** Every edge $\vec{f}$ has one or two switches in $c_0$: $1 \leq |S_{c_0}(\vec{f})| \leq 2$.




- X1 gives $|S_{c_0}(\vec{f})| \in \{1, 2\}$ for every edge.

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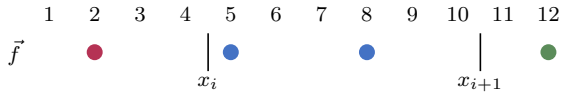
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- Start with colour 0 and advance $0 \to 1 \to 2 \to 0$ at each switch.
- $c_3(f_4) - c_3(f_1) \equiv |S_{c_0}(\vec{f})| \pmod 3$, and $|S_{c_0}(\vec{f})| \in \{1, 2\}$. Thus no edge is monochromatic.

Thank you!